



UNIVERSITY OF BUEA

TASK

6

CULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER
ENGINEERING

**CEF 440: INTERNET PROGRAMMING AND MOBILE
PRORAMMING**

**TASK6: DATABASE DESIGN AND
IMPLEMENTATION**

GROUP 19

**Design and Implementation of a Mobile App for Collection of User
Experience Data from Mobile Network Subscribers**

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1 Introduction

The Quality of Experience (QoE) Network Monitoring System is designed to collect, store, analyze, and display network performance data obtained from users and monitoring services. The database serves as the central repository for storing network measurements, user information, QoE metrics, and generated reports.

The objective of this task is to design and implement a reliable database structure that supports data collection, processing, visualization, and reporting functionalities of the system.

2 Data Elements

The first step in database design was identifying all the data required by the system.

Main Data Entities

1. Users

Stores information about users accessing the platform.

Attribute	Data Type	Description
user_id	INT	Unique identifier
name	VARCHAR(100)	User name
email	VARCHAR(100)	User email
password	VARCHAR(255)	Encrypted password
role	VARCHAR(20)	User role
created_at	TIMESTAMP	Registration date

2. Network Measurements

Stores network performance metrics collected from devices.

Attribute	Data Type	Description
measurement_id	INT	Unique identifier
user_id	INT	References User
latency	FLOAT	Network latency (ms)
packet_loss	FLOAT	Packet loss (%)
jitter	FLOAT	Jitter value
download_speed	FLOAT	Download speed (Mbps)
upload_speed	FLOAT	Upload speed (Mbps)
location	VARCHAR(255)	Measurement location
recorded_at	TIMESTAMP	Date collected

3. QoE Scores

Stores calculated Quality of Experience scores.

Attribute	Data Type	Description
qoe_id	INT	Unique identifier
measurement_id	INT	Related measurement
qoe_score	FLOAT	Calculated QoE value
rating	VARCHAR(20)	Excellent/Good/Fair/Poor
generated_at	TIMESTAMP	Date generated

4. Reports

Stores generated system reports.

Attribute	Data Type	Description
report_id	INT	Unique identifier
user_id	INT	Generated by user
title	VARCHAR(100)	Report title
description	TEXT	Report details
created_at	TIMESTAMP	Creation date

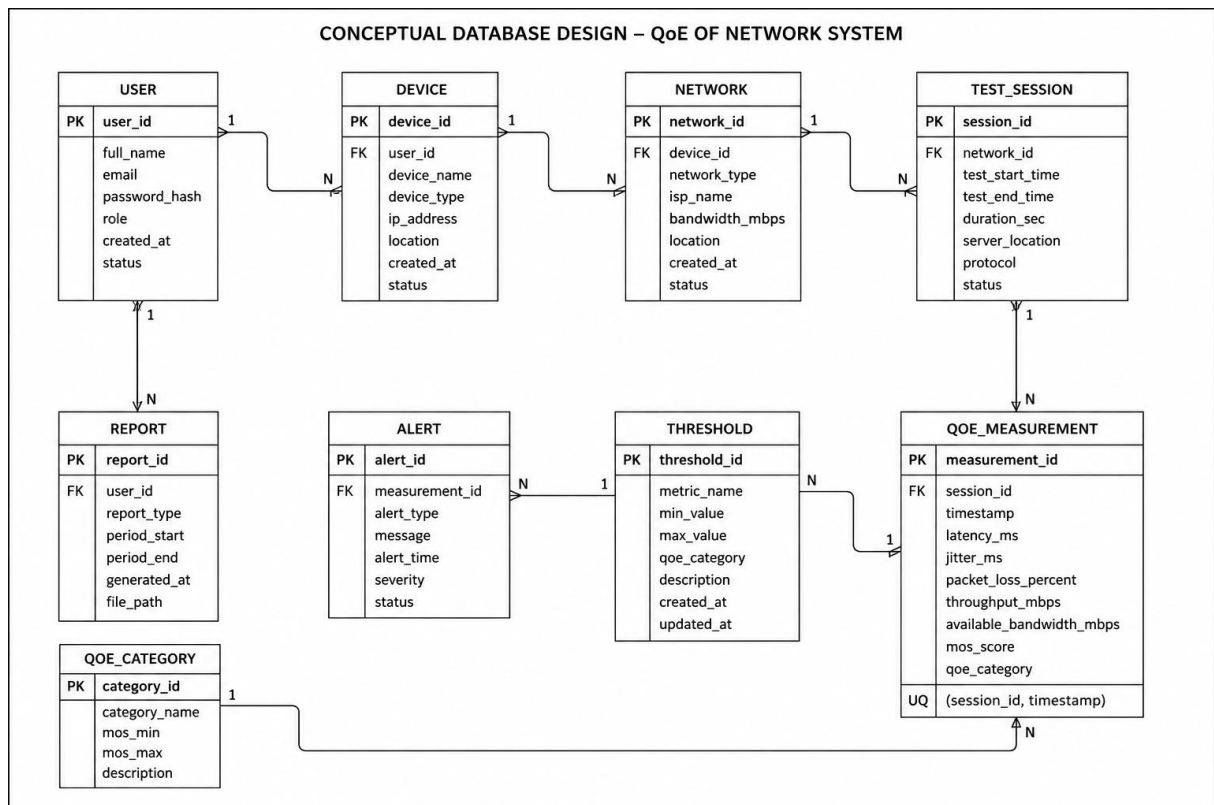
3 Conceptual Design

The conceptual design was created to represent the relationships between entities before implementation.

Relationship Analysis

1. One User can create many Network Measurements.
2. One Network Measurement produces one QoE Score.
3. One User can generate many Reports.
4. QoE Scores depend on Network Measurements.

Conceptual Model: The conceptual design ensures data consistency and eliminates redundancy.



4 Entity Relationship Diagram (ERD)

The Entity Relationship (ER) diagram describes in the document ENTITY RELATIONSHIP (ER) DIAGRAM REPORT.docx. The ER diagram serves as the blueprint for a relational database supporting a Quality of Experience (QoE) mobile application.

The database is designed to:

- Collect objective network performance metrics
- Collect subjective user feedback
- Support analytical reporting
- Enable future AI-based QoE prediction

Entities in the Database

The database contains nine entities:

Entity Description

User Mobile network subscribers using the application

Session Monitoring sessions initiated by users

Network Performance Objective QoS measurements (speed, latency, signal, etc.)

User Feedback Subjective user ratings and comments

Location Geographical information for measurements

Notification Surveys and feedback requests sent to users

Alert Incidents generated when performance falls below thresholds

Analytics Report Aggregated statistics for dashboards

QoE Prediction AI-generated satisfaction predictions

Relationships

All relationships in this ER diagram are one-to-many (1:M). From Entity to Entity
Relationship Description

- 1 .User Session One user → many sessions
- 2 .User Network Performance One user → many performance records
- 3 .User User Feedback One user → many feedback entries
- 4 .User Notification One user → many notifications

5 .Session Network Performance One session → many measurements

6 .Session User Feedback One session → many feedback entries

7 .Location Network Performance One location → many measurements

8 .Location User Feedback One location → many feedback entries

9 .Network Performance Alert One performance record → many alerts

Derived Entities (No Direct Relationships Shown)

- Analytics Report – Generated from operational data
- QoE Prediction – Generated from AI models

Primary and Foreign Keys

- Primary Keys (Uniquely identify each record)
 - user_id
 - session_id
 - record_id (Network Performance)
 - feedback_id (User Feedback)
 - location_id
 - notification_id
 - alert_id

- Foreign Keys (Establish relationships)

Foreign Key Located In References

user_id Session User

user_id Network Performance User

user_id User Feedback User

user_id Notification User

session_id Network Performance Session

session_id User Feedback Session

location_id Network Performance Location

location_id User Feedback Location

performance_id Alert Network Performance

Database Normalization

The database design satisfies normalization principles:

- No duplicate data
- Independent entities are separated
- Related information stored in dedicated tables
- Referential integrity maintained via foreign keys

Advantages of the Design:

Advantage Description

Efficient storage Optimized for QoE measurements

Reduced redundancy Normalized structure

Data integrity Enforced via foreign keys

Easy retrieval Clear entity relationships

Geographical analysis Location data integrated

Performance monitoring Alerts tied to metrics

Subjective + objective Both data types captured

Scalability Supports future enhancements

AI ready Supports ML-based prediction

ER Diagram Notation

Symbol Meaning

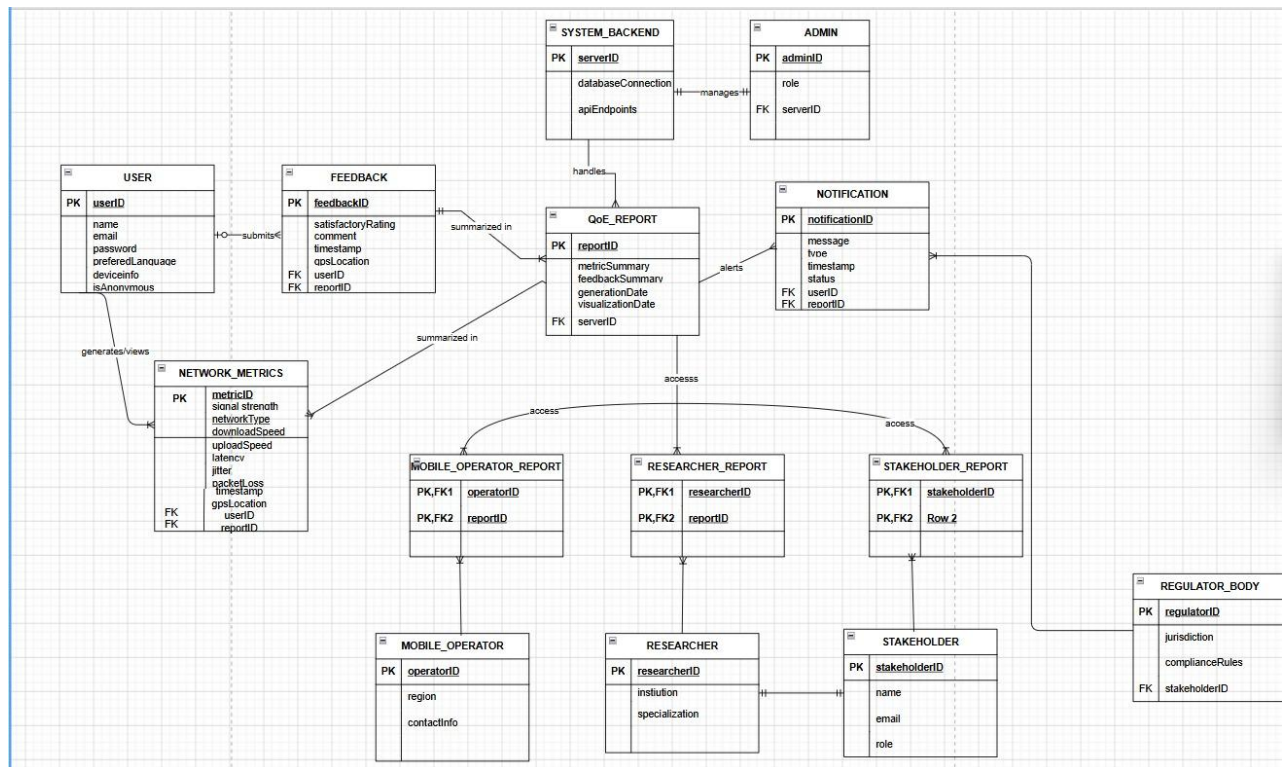
Rectangle Entity (table)

Key symbol Primary Key

Connecting line Relationship

Crow's foot (M) Many side of 1:M relationship

Diagram:



The ER diagram provides a normalized, scalable database design suitable for implementation in MySQL Workbench. It supports:

- Core QoE data collection from mobile users
- Session-based monitoring
- Integration of objective metrics and subjective feedback
- Location-aware analysis
- Automated alerting
- Future AI-based QoE prediction

This design meets the requirements for the proposed QoE mobile application.

5 Database Implementation

The database was implemented using MySQL.

Database Name

qoe_monitoring_db

Creating the Database

```
CREATE DATABASE qoe_monitoring_db;
USE qoe_monitoring_db;
```

Creating Users Table

```
CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100) UNIQUE,
  password VARCHAR(255),
  role VARCHAR(20),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Creating Network Measurements Table

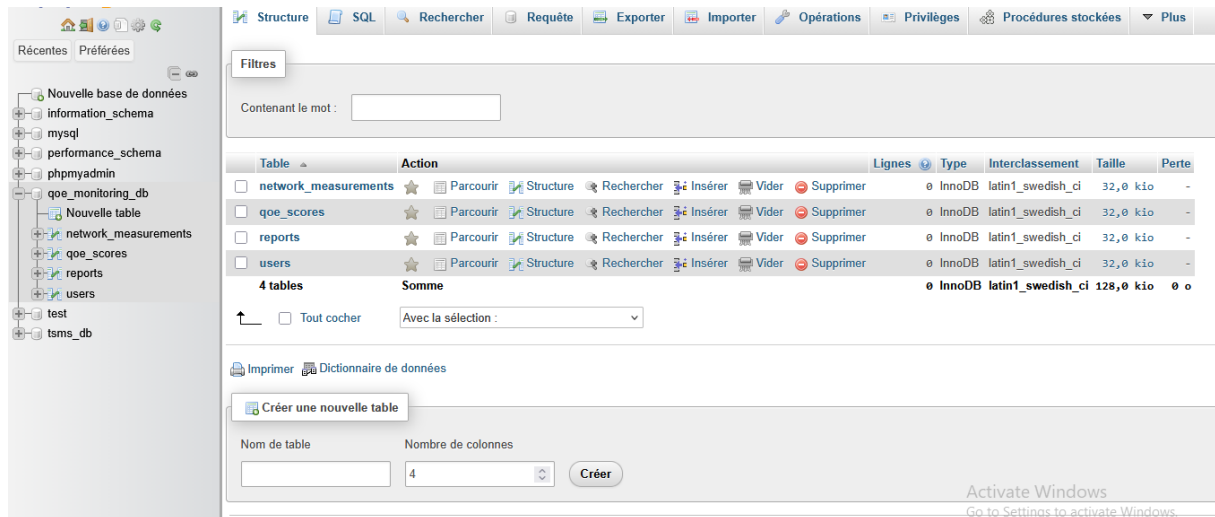
```
CREATE TABLE network_measurements (
  measurement_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  latency FLOAT,
  packet_loss FLOAT,
  jitter FLOAT,
  download_speed FLOAT,
  upload_speed FLOAT,
  location VARCHAR(255),
  recorded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id)
    REFERENCES users(user_id)
);
```

Creating QoE Scores Table

```
CREATE TABLE qoe_scores (
  qoe_id INT AUTO_INCREMENT PRIMARY KEY,
  measurement_id INT,
  qoe_score FLOAT,
  rating VARCHAR(20),
  generated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (measurement_id)
    REFERENCES network_measurements(measurement_id)
);
```

Creating Reports Table

```
CREATE TABLE reports (
  report_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  title VARCHAR(100),
  description TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id)
    REFERENCES users(user_id)
);
```



6 Backend Implementation

The backend was developed using Node.js and Express.js to handle communication between the frontend and the database.

Backend Responsibilities

- User authentication
- Data collection
- QoE score computation
- Report generation
- API management

Database Connection Configuration

```
const mysql = require('mysql2');
```

```
const db = mysql.createConnection({
  host: 'localhost',
  user: 'root',
  password: '',
  database: 'qoe_monitoring_db'
});
```

```
module.exports = db;
```

The backend exposes RESTful APIs that allow the frontend dashboard to retrieve network performance data and QoE statistics.



Step 1: Install Dependencies

A database connection file was created inside the config folder.

Step 4: Retrieve Data

The API sends JSON responses to the React Native frontend where charts and dashboards display network statistics in real time.



8 Challenges Encountered

Challenge	Solution
Foreign key errors	Verified table relationships
Database connection failures	Corrected credentials and ports
API response delays	Optimized queries
Data duplication	Added constraints and validation
Backend to database connection	Multiple trials

9 Conclusion

The database for the QoE Network Monitoring System was successfully designed and implemented using MySQL. The conceptual design and ERD provided a clear structure for development, while the backend integration enabled seamless communication between the application and the database. The implemented database supports efficient storage, retrieval, and analysis of network performance data, making it a critical component of the overall system.